Dual Lock-In Amplifier

SR835 — DSP lock-in amplifier

- Synchronous sampling architecture
- Spur-free sine output
- Phase-linear FIR filters
- Dual-channel demodulation
- 1 mHz to 1 MHz frequency range
- >100 dB dynamic reserve
- Time constant from 1 μs to 1 day
- 0.01 degree phase resolution
- USB and Ethernet interfaces

- SR835 … $3999 (U.S. list)

The SR835 is built on a fundamentally new architecture: synchronous sampling. A fractional-N clock synthesizer with sub-picosecond jitter locks the ADC and DDS clocks to an exact integer multiple of the reference frequency. This single innovation eliminates DDS spurs, enables synchronous filtering at any frequency, and allows time constants as short as one signal period—all without the compromises of traditional lock-in designs.

The result is a dual-channel, dual-phase lock-in amplifier covering 1 mHz to 1 MHz in a single 2U rack instrument. A dedicated 1 GHz DSP with a hardware FIR accelerator provides phase-linear digital filters that preserve signal waveshape with no overshoot and finite settling time. Two independent measurement channels share a common oscillator, enabling simultaneous two-signal measurement, harmonic detection, or vector network analysis—all without external hardware. Live Ethernet streaming, a built-in scan engine, FFT display, and a unified SCPI command set complete the instrument.

Synchronous Sampling

Traditional lock-in amplifiers run their ADC and DDS clocks independently of the signal frequency. The SR835 takes a different approach: a fractional-N clock synthesizer with sub-picosecond jitter locks both clocks to an exact integer multiple of the reference. Because every sample window aligns precisely with the signal period, a simple boxcar sum over one cycle completely nulls the ω and 2ω mixer products—no high-order filter required. This is the foundation from which the SR835's other advantages follow.

Spur-Free Sine Output

In a conventional DDS, the output frequency is not an exact sub-multiple of the clock, producing phase-truncation spurs that can reach –70 dBc. With synchronous sampling, the DDS clock is an integer multiple of the output frequency, so every cycle of the waveform is identical. The spurs collapse underneath the fundamental and the output is spectrally clean. The DDS also supports arbitrary waveform generation for stimulus applications that require non-sinusoidal drive.

Phase-Linear Digital Filters

The SR835 replaces the cascaded RC (IIR) filters of earlier lock-in amplifiers with phase-linear FIR low-pass filters, executed on a 1 GHz DSP with a dedicated hardware FIR accelerator. These filters have equal group delay at all frequencies: the output preserves the true time-domain shape of the input signal with no waveform distortion, no overshoot, and an impulse response that settles to exactly zero—eliminating the long exponential tails of RC filters. For control-loop applications, the reduced latency of a short FIR is a direct improvement over an equivalent-bandwidth IIR cascade.

Synchronous Filter

Because sampling is synchronous, a boxcar FIR that sums over exactly one signal period acts as a perfect comb filter, placing nulls at ω and every harmonic. Unlike older sync filters that worked only at low frequencies, this provides ≥ 80 dB rejection of mixer products across the full 1 mHz to 1 MHz range. The filter tracks the reference automatically, and also enables true DC measurement of the signal by notching out the ω component. The fastest time constant equals one period of the signal.

Auto Reference Tracking

A slow digital phase-locked loop continuously tracks the user's signal frequency, keeping the internal oscillator locked even when the reference drifts. This removes the need for a clean external reference in applications where the signal source is not frequency-stable.

Dual-Phase Demodulation

Each channel computes X and Y simultaneously by multiplying the input against precise cosine and sine waves derived from a numerical oscillator. No square-wave switching or hidden latency—just true quadrature outputs with stable phase.

Redesigned Inputs

A fully differential input topology features a precision JFET front-end stage with 2 nV/$\sqrt{\text{Hz}}$ input noise and an input impedance of 10 MΩ. Inputs are fully floating, with an optional chassis ground connection. A rear-panel single-ended input provides 1 nV/$\sqrt{\text{Hz}}$ noise for applications demanding the lowest possible noise floor.

Inputs are fully protected against overvoltage events by means of bootstrapped transient voltage suppressors. In case of sustained overload, damage is avoided by automatically disconnecting the inputs. The "trip" condition can be cleared from the front panel or via the remote interface.

Time Constant and Filter Slope

The time constant is continuously adjustable from one signal period (or 1 μs) up to a full day, or stepped in 1–2–5 decades. Filter slope is selectable from 6 to 24 dB/octave. Coefficients are precomputed and swapped seamlessly, so bandwidth changes don't introduce glitches or transients. An up-sampler interpolates between output samples, eliminating staircase artifacts at long time constants.

Harmonic Detection

Higher harmonic detection is built into the reference generation. By scaling the internal phase accumulator, the instrument locks to any integer multiple from 1× to 99×, within range. Each channel can set its harmonic independently, allowing simultaneous measurement of different spectral components.

Dual-Reference Detection

With two references, the instrument locks to their difference frequency. The internal oscillator and external reference are both tracked, and the demodulator measures the intermodulation product at their offset. This enables two-tone measurements without external mixers or added hardware.

Vector Network Analysis

The two independent measurement channels can be configured as a ratio measurement: one channel monitors the stimulus, the other the response. This turns the SR835 into a vector network analyzer for characterizing transfer functions

with full amplitude and phase information, without additional instruments.

Auxiliary Output Measurement Mode

The auxiliary outputs can be driven by any measured quantity. The selected signal is normalized, offset, scaled, and mapped into the output range with configurable full-scale, offset, and expansion. This allows zooming in on small variations or centering signals, with clipping detection indicating when limits are exceeded.

Parameter Scan Engine

Sweeps are handled internally, varying one parameter at a time in a controlled way. Frequency, amplitude, offsets, or auxiliary outputs can be ramped between limits over a set duration, using linear or logarithmic spacing. Scans can run once, repeat, or reverse direction, all while measurement and capture continue uninterrupted.

Data Capture Buffer

Data is written into onboard memory, supporting sustained acquisition from very slow trends to high-speed sampling. The buffer holds up to one million points and supports multiple formats, from X traces to full vector data. It can run as single-shot, circular, or logarithmic capture, and can be externally triggered.

Real-Time Ethernet or USB Streaming

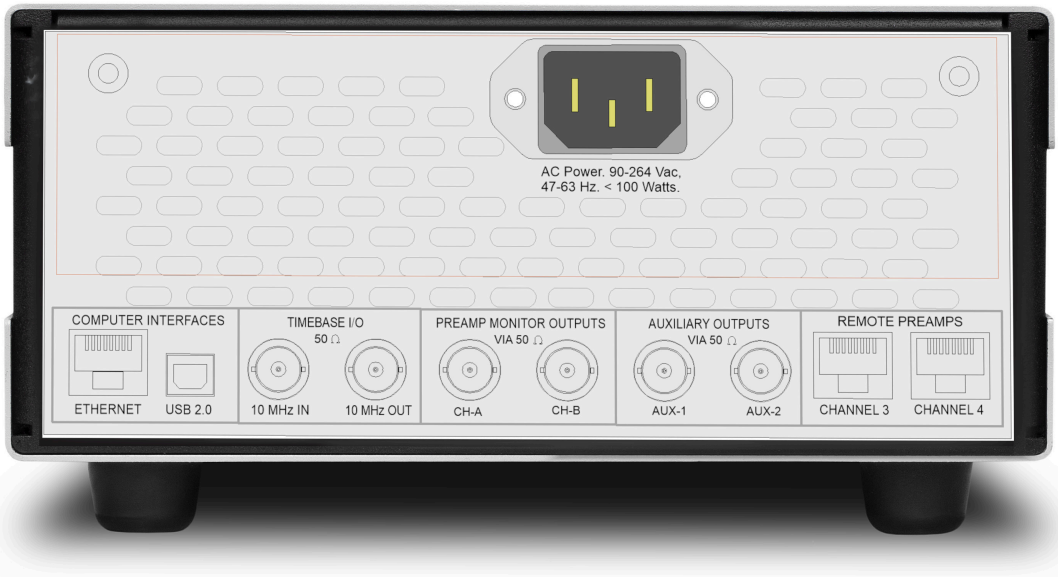
Demodulated data can be streamed continuously over Ethernet or USB interfaces for real-time use. Packets are sent using standard protocols, with selectable formats and sizes to balance throughput and latency. Streaming runs independently of capture, and sequence numbers allow detection of dropped packets.

Front-Panel Touchscreen GUI

All functionality is exposed through a modern touchscreen interface that replaces layered menus and knobs. Parameters are organized hierarchically with touch targets sized for reliable use. Physical increment buttons remain for fine control, giving both quick navigation and precise adjustment.

Ordering Information

SR835	Dual-Channel Lock-In Amplifier	$3999
SR801	Remote Voltage Preamplifier (100 MΩ, 1.2 nV/√Hz)	$799
SR802	Remote Current Preamplifier	$799
O835RMD	Double rack mount kit	$100
O835RMS	Single rack mount kit	$100



Signal Input

Specification	Value
Input configuration	Single-ended (A), differential (A-B), or current (I)
Input impedance, voltage (DC coupled)	10 MOhm ‖ 25 pF
Input impedance, voltage (AC coupled)	10 MOhm ‖ 25 pF; high-pass f0 = 0.16 Hz
Current input impedance to virtual ground	1 kOhm
Current transimpedance gain options	100 kOhm, 1 MOhm, 10 MOhm, 100 MOhm
Current input bandwidth (100k / 1M / 10M / 100M)	3 MHz / 70 kHz / 7 kHz / 700 Hz
Input voltage noise	2.5 nV/sqrt(Hz) at 1 kHz (gain x1, typical)
Input current noise (1 MOhm / 100 MOhm)	130 fA/sqrt(Hz) / 13 fA/sqrt(Hz) at 1 kHz
CMRR (DC coupled)	>100 dB at DC–10 kHz; −6 dB/octave above 10 kHz
Gain accuracy	+/-1% below 100 kHz; +/-2% to 1 MHz
Gain stability	<50 ppm/deg C
Harmonic distortion	<−80 dBc below 100 kHz; <−70 dBc above 100 kHz
Preamp gain steps	1, 2, 4, 8, 16, 32, 64, 128, 256 (binary, relay-switched)
Line notch attenuation	>=30 dB at 50/60 Hz and 100/120 Hz
Bandpass filter Q	5, auto-tracking

Reference Channel

Specification	Value
Internal frequency range	1 mHz to 1 MHz
Internal frequency accuracy	25 ppm + 30 uHz
Internal frequency resolution	6 digits or 0.1 mHz (whichever is greater)
External reference frequency range	0.5 Hz to 1 MHz (sine or TTL)
External reference, minimum sine amplitude	400 mV peak-to-peak
External reference input impedance	1 MOhm (fixed)
PLL acquisition time	max(2 cycles + 5 ms, 40 ms)
Phase offset range	+/-360 000 deg (displayed wrapped to +/-180 deg)
Phase resolution	360/2^32 deg = 8.4 x 10^−8 deg
Phase noise (internal reference)	<0.0001 deg rms at 1 kHz (100 ms TC, 12 dB/oct)
Phase noise (external TTL reference)	<0.001 deg rms at 1 kHz (100 ms TC, 12 dB/oct)
Phase drift below 20 kHz	<0.002 deg/deg C
Phase drift below 200 kHz	<0.02 deg/deg C
Phase drift below 1 MHz	<0.2 deg/deg C
Orthogonality (X vs. Y)	90 deg +/- 0.001 deg
Absolute phase error	<1 deg above 10 Hz
Harmonic detection range	N = 1 to 99; N x f <= 1 MHz

Demodulator and Filters

Specification	Value
Harmonic rejection	−80 dBc
Time constant range	1 us to 86 400 s; any value, 1-2-5 per decade stepping
Filter slopes	6, 12, 18, 24 dB/octave
ENBW—6 dB/oct	1/(4T)
ENBW—12 dB/oct	1/(8T)
ENBW—18 dB/oct	3/(32T)
ENBW—24 dB/oct	5/(64T)
Settling to 99% (6/12/18/24 dB/oct)	5T / 7T / 9T / 10T
Synchronous filter rejection	>=80 dB at N x f_det; always active, full range
DC output stability (digital)	No drift
DC output stability (Aux DAC)	<5 ppm/deg C

Sine Output

Specification	Value
Frequency range	1 mHz to 1 MHz
Amplitude range	1 nVrms to 2 Vrms (differential, open circuit)
Amplitude into 50 Ohm	Half open-circuit value
Amplitude accuracy	+/-1%
Amplitude stability	<50 ppm/deg C
Harmonic distortion	<−80 dBc below 100 kHz; <−70 dBc above 100 kHz
Output impedance	50 Ohm (differential)
DC offset range	+/-5 V
DC offset resolution	0.1 mV

Maximum output (| DC| + V_peak)	+/-6 V
Output anti-alias filter	7th-order Cauer LPF, Fc = 9 MHz; −100 dB by 19 MHz

Auxiliary Outputs

Specification	Value
Number of channels	2 (Aux-1, Aux-2)
Voltage range	+/-10.500 V
Resolution	1 mV (12-bit DAC)
Output impedance	<50 Ohm
Output current limit	10 mA

Remote Interfaces

Specification	Value
USB	USB 2.0 device, USBTMC class
Ethernet	100BASE-T; VXI-11 + TCP socket (default port 5025)
Command format	SCPI (IEEE 488.2 syntax)
Input buffer	256 characters
Output buffer	256 characters
Non-volatile setup slots	9
Maximum streaming rate	1.25 MHz (Ethernet)
Streaming data formats	Float32 (4 B/sample), Int16 (2 B/sample)
Trigger-to-first-sample latency	<=2 ms

10 MHz Timebase

Specification	Value
Timebase In	1 Vrms, 10 MHz; input impedance 50 Ohm
Timebase Out	1 Vrms, 10 MHz; output impedance 50 Ohm
Phase-lock range	+/-50 ppm pull-in range
Multi-unit synchronization	<1 ps rms (typical) of shared 10 MHz

General

Specification	Value
Mains input voltage	100 / 120 / 220 / 240 VAC +/-10%, switch-selectable
Mains frequency	50 or 60 Hz
Power consumption	60 W typical
Fuse (100/120 V)	1 A, 3AG slow-blow
Fuse (220/240 V)	0.5 A, 3AG slow-blow
Operating temperature	+10 to +40 deg C; specs apply +18 to +28 deg C
Storage temperature	−25 to +65 deg C
Operating humidity	<90% non-condensing
Storage humidity	<95% non-condensing
Form factor	19″ rack, 2U (bench use with handles)
Width	432 mm (17″)
Height	133 mm (5.25″)
Depth	432 mm (17″)
Weight	12 kg (26 lbs)